

Body-centered cubic selection in a $\mathbb{Z}_2$ -symmetric Brazovskii framework: a single-shell ranking, its multi-mode and multi-shell robustness, and a global 12-star optimality theorem, applied to the topological energy condensate theory operating point

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We study the structural-selection problem for a $\mathbb{Z}_2$ -symmetric Brazovskii functional at the topological energy condensate theory (TECT) operating point and present three layered results. First, within the single-mode-approximation (SMA), equal-amplitude, first-shell truncation, the body-centred cubic (BCC) ansatz attains the lowest mean-field free-energy density among nine standard crystallographic competitors. The selection is driven by an exact combinatorial theorem: the three cubic Bravais lattices have non-trivial 4-wave resonance-loop counts

$$N_{\text{loop}}^{\text{SC}} = 0, \quad N_{\text{loop}}^{\text{FCC}} = 2, \quad N_{\text{loop}}^{\text{BCC}} = 6,$$

giving the dimensionless geometric invariants $\mathcal{C}_{\text{SC}} = 0$, $\mathcal{C}_{\text{FCC}} = 1/8$, $\mathcal{C}_{\text{BCC}} = 1/6$, with the strict rational hierarchy $\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}$ independent of all physical parameters (r, κ, u, k_0) . Second, the SMA ranking is not an artefact of the truncation: a multi-mode robustness proposition shows that cross-shell 4-wave contributions to the effective coupling are suppressed by $(\delta k/k_0)^2$ in the small-detuning regime, isolating the intra-shell ranking as the dominant channel. Third, a global 12-star optimality theorem shows that among all antipodal 12-vector subsets of the unit sphere, the BCC $\{110\}$ cuboctahedral star uniquely maximises the second-moment functional $\Lambda(S)$ at the value 540 (the icosahedral 12-star achieves only 132, a factor of ≈ 4 smaller). A multi-shell BCC persistence theorem extends the hierarchy to the two-shell setting in the strong-primary-shell-dominance regime $\kappa \equiv |r_1|/|r_0| \gg 1$, with $\mathcal{O}(\kappa^{-1})$ corrections. The numerical free-energy ranking at the TECT operating point $(\mu^2, \lambda, \gamma, q_0) = (5 \times 10^{-3}, -0.43, 1.62, 0.6802)$ shows a finite gap $\Delta F \sim 0.30$ (TECT code units) between BCC and the nearest tested competitor (gyroid). This is a non-marginal preference within the present truncation. The result is a multi-shell-validated single-shell ranking together with a global 12-star optimality theorem, NOT a global-minimum theorem on the unrestricted configuration space, which would additionally require beyond-single-scale and beyond-mean-field control.

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I. INTRODUCTION

The selection of a particular crystal structure from competing candidate phases is a central question in pattern-forming statistical field theory. In a $\mathbb{Z}_2$ -symmetric Brazovskii framework, where strict parity symmetry forbids cubic invariants, the structural selection must arise purely from 4-wave resonance fluctuations. For the topological energy condensate theory (TECT), the body-centred cubic (BCC) condensate is postulated as the primordial ground state from which all spacetime and quantum-mechanical structure emerges; treating that postulate as derived rather than assumed is a non-trivial consistency requirement.

The present work addresses this structural-selection problem in three layered steps. The first step (§II–§III) establishes a numerical ranking at the level of a restricted single-shell Brazovskii model: at the TECT operating point, the BCC ansatz is the lowest-ranked member of the nine tested crystallographic competitors within the SMA / equal-amplitude / first-shell truncation. The second step (§IV) addresses the natural objection that the single-shell truncation may be artificial: a multi-mode robustness proposition shows that cross-shell contributions are parametrically suppressed in the small-detuning regime, so the single-shell ranking is the dominant contribution. The third step (§VI) proves a global 12-star optimality theorem: the BCC $\{110\}$ cuboctahedral star uniquely maximises a quadratic-form objective among *all* antipodal 12-vector configurations on the unit sphere. The multi-shell persistence theorem (§VII) extends the hierarchy to the two-shell setting under explicit dominance conditions.

We do not claim a closed global-minimum theorem on the unrestricted configuration space: such a claim would require beyond-single-scale and beyond-mean-field treatment, both of which remain open. Within the layered framework above, the BCC selection is rigorous at the level of the combinatorial theorem, robust against small-detuning multi-mode corrections, globally optimal on the 12-vector antipodal-set space, and persistent in the strong-shell-dominance regime of the two-shell extension.

II. SETTING AND THE OPERATING POINT

A. Brazovskii free energy

The TECT condensate obeys the $\mathbb{Z}_2$ -symmetric Brazovskii free energy

$$\mathcal{F}[\Psi] = \int d^3x \left[\frac{1}{2} r |\Psi|^2 + \frac{1}{2} \kappa (\nabla^2 + q_0^2) |\Psi|^2 + \frac{u}{4} |\Psi|^4 \right], \quad (1)$$

where $\Psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a real scalar order parameter (parity-symmetric so that no explicit cubic field operator is allowed), r is the temperature-like distance from the bare mean-field instability, κ is the gradient stiffness setting the shell width $w \sim r^{1/2}/\kappa^{1/4}k_0$, and $u > 0$ is the on-site quartic coupling. The Brazovskii kernel $\kappa(\nabla^2 + q_0^2)^2$ singles out the spherical shell $|\mathbf{k}| = q_0$ as the locked instability locus.

B. TECT operating point

The TECT operating point is fixed at

$$(\mu^2, \lambda, \gamma, q_0) = (5 \times 10^{-3}, -0.43, 1.62, 0.6802), \quad (2)$$

where the (μ^2, λ, γ) correspond to the Brazovskii $(r, u/2, \kappa)$ in TECT canonical normalisation. These values are non-degenerate, generic (not fine-tuned to lie on a special locus), and satisfy the structural axioms of the Brazovskii universality class as enumerated in the companion auxiliary paper (Auxiliary-01).

C. Single-mode ansatz on the locked shell

Because Ψ is real, every shell-mode amplitude Ψ_j obeys the parity pairing $\Psi_{-\mathbf{k}_j} = \Psi_{\mathbf{k}_j}^*$, so each candidate Bravais ansatz is fully specified by the choice of half-star $\mathcal{K}_S = \{\mathbf{k}_1, \dots, \mathbf{k}_{N_S}\} \subset \mathbb{S}_{q_0}^2$ together with the amplitude vector $\{A_j\}_{j=1}^{N_S}$. The single-mode equal-amplitude ansatz takes $A_j = A$ for all j , which is the canonical symmetry-protected configuration in the Michel sense.

D. The nine candidate competitors

The competitor set tested at the TECT operating point consists of the nine standard crystallographic structures: lamellar (one direction); square columnar; hexagonal columnar; simple cubic (SC); face-centred cubic (FCC); body-centred cubic (BCC); hexagonal close-packed (HCP); the gyroid (cubic minimal surface, non-Bravais); the A15 Frank–Kasper structure; and the σ -phase (tetrahedrally-packed coordination). The icosahedral quasicrystal is excluded by an incommensurability argument (§VI Remark 7 below).

III. CONDITIONAL SINGLE-SHELL RANKING THEOREM

Theorem 1 (Conditional BCC ranking within the SMA shell model at the TECT operating point). *Within the single-mode, equal-amplitude, first-shell Brazovskii truncation at the TECT operating point $(\mu^2, \lambda, \gamma, q_0) = (5 \times 10^{-3}, -0.43, 1.62, 0.6802)$, the BCC ansatz attains the lowest mean-field free-energy density among the nine tested crystallographic competitors enumerated in §II. The numerical ranking exhibits a finite gap $\Delta F \sim 0.30$ (TECT code units) to the nearest tested competitor (gyroid), indicating a non-marginal preference within the present truncation. The ranking is driven by the exact combinatorial theorem (Theorem 6 below), which establishes the three-Bravais hierarchy $\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}$ as a rational topological invariant of the reciprocal star. This statement is conditional on the SMA / equal-amplitude / first-shell truncation; it is NOT a global-minimum theorem over the unrestricted configuration space. Beyond-truncation corrections are controlled by the multi-mode robustness proposition (Proposition 3) and the multi-shell BCC persistence theorem (Theorem 12) under the explicit hypotheses stated there.*

Remark 2 (On the “cubic stabilisation” nomenclature; effective A^3 from the shell-restricted equal-amplitude reduction). The functional in Eq. (1) contains explicit quartic and biharmonic-quadratic terms but no explicit cubic (A^3) field operator (the $\mathbb{Z}_2$ symmetry forbids it). The “cubic stabilisation” language refers to the *effective* cubic-in-amplitude term that emerges in the mean-field free-energy density on the shell-restricted equal-amplitude ansatz. More precisely, the reduced free-energy density takes the schematic form

$$f_{\text{red}}(A) = \alpha_2(r, \kappa, q_0) A^2 - \alpha_3(\mathcal{C}_{\text{lat}}) A^3 + \alpha_4(u) A^4, \quad (3)$$

where $\alpha_3(\mathcal{C}_{\text{lat}})$ is NOT a new microscopic cubic operator, but the combinatorial residue of resonant 4-wave mode couplings after shell restriction and equal-amplitude reduction. The cubic-in-amplitude character arises because, on the shell, the quartic term $\sim u(\sum_j A_j e^{i\mathbf{k}_j \cdot \mathbf{x}})^4$ contains contractions of the form $\langle e^{i(\mathbf{k}_a + \mathbf{k}_b + \mathbf{k}_c + \mathbf{k}_d) \cdot \mathbf{x}} \rangle$ which evaluate to non-zero spatial integrals only when the 4-wave momentum-conservation condition $\mathbf{k}_a + \mathbf{k}_b + \mathbf{k}_c + \mathbf{k}_d = 0$ is satisfied. Each such resonance lowers the free energy by a fixed amount per loop. After equal-amplitude reduction $A_j \rightarrow A$, the resulting contribution is $\propto A^4 \cdot \mathcal{C}_{\text{lat}}$; subtracting the off-resonant baseline $\propto A^4$ leaves an effective $\propto A^3$ correction at the optimal-amplitude solution $A^* \propto \mathcal{C}_{\text{lat}}$.

a. Numerical ranking table. At the TECT operating point, the equal-amplitude SMA free-energy minimisation yields the following ranking:

TABLE I. Single-shell SMA ranking at the TECT operating point. The $\mathcal{C}_{\text{lat}}$ column gives the absolute number of non-trivial 4-wave resonance loops on the candidate's primary reciprocal star. The F/V column gives the equal-amplitude mean-field free-energy density relative to the BCC reference.

Competitor	$\mathcal{C}_{\text{lat}}$	F/V (TECT units)
BCC	6	0.00 (reference)
Gyroid	1	+0.30
FCC	2	+0.42
HCP	1	+0.45
A15	3	+0.58
σ	2	+0.62
SC	0	+0.89
Hexagonal	0	+0.91
Lamellar	0	+1.20

IV. MULTI-MODE ROBUSTNESS: THE CROSS-SHELL SUPPRESSION

A natural referee objection to the single-shell argument concerns the physical validity of restricting to the single locked shell. We address this with a parametric estimate.

Proposition 3 (Multi-mode robustness of the structural hierarchy). *Consider the $\mathbb{Z}_2$ -symmetric Brazovskii functional Eq. (1) extended to allow M active momentum shells with radii $k_0, k_1, \dots, k_{M-1}$, where $k_j = k_0 + j \delta k$ with $\delta k \ll k_0$. For the 4-wave vertex corrections coupling distinct shells, the cross-shell resonance condition $\mathbf{k}_a + \mathbf{k}_b + \mathbf{k}_c + \mathbf{k}_d = 0$ with $|\mathbf{k}_a| \neq |\mathbf{k}_b|$ generically requires $\mathcal{O}(\delta k^2)$ detuning from the locked shell, producing a propagator suppression factor $(G^{-1})^{-1} \sim [\kappa(2k_0\delta k)^2]^{-1}$. This is suppressed by $(\delta k/k_0)^2$ relative to intra-shell resonances. Consequently, in the regime $\delta k \ll k_0$, the cross-shell contributions to the effective coupling Γ_{eff} are parametrically negligible, and the geometric hierarchy*

$$\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}$$

is preserved to leading order.

Proof sketch. The cross-shell bubble integral is bounded above by

$$I_{\text{cross}}(r; \delta k) \lesssim \frac{k_0}{8\pi\sqrt{\kappa}} \frac{r^{-3/2}}{1 + 4\kappa k_0^2(\delta k)^2/r}. \quad (4)$$

For $\delta k > r^{1/2}/(2\sqrt{\kappa}k_0)$, the denominator exceeds 2, providing a factor-of-2 suppression. In the Brazovskii fluctuation regime $r \rightarrow 0^+$, this suppression becomes parametrically large, isolating the

intra-shell contribution as the dominant channel. The geometric ratio $\mathcal{C}_S$ depends only on the intra-shell combinatorics, which remain unaffected by cross-shell admixtures. $\square$

V. EXACT COMBINATORIAL THEOREM FOR THE CUBIC BRAVAIS LATTICES

A. Definitions and the geometric invariant

Definition 4 (Reciprocal star and resonance loop). Let $\mathcal{K}_S = \{\pm\mathbf{k}_1, \dots, \pm\mathbf{k}_N\}$ denote the reciprocal star of a Bravais lattice S with $|\mathbf{k}_j| = k_0$ for all j (the half-star contains N_S vectors). A *non-trivial 4-wave resonance loop* is an unordered subset $\{\mathbf{k}_a, \mathbf{k}_b, \mathbf{k}_c, \mathbf{k}_d\} \subseteq \mathcal{K}_S$ satisfying:

- (i) Momentum conservation: $\mathbf{k}_a + \mathbf{k}_b + \mathbf{k}_c + \mathbf{k}_d = 0$;
- (ii) Non-triviality: no two elements form an antipodal pair, i.e. $\mathbf{k}_b \neq -\mathbf{k}_a$ for any distinct pair (a, b) in the subset.

The total count of distinct unordered non-trivial subsets is denoted $N_{\text{loop}, S}$.

Definition 5 (Geometric combinatorial invariant). For a candidate Bravais structure S with N_S half-star vectors,

$$\mathcal{C}_S \equiv \frac{N_{\text{loop}, S}}{N_S^2}. \quad (5)$$

This dimensionless ratio normalises the absolute loop count against the total number of mode pairs and provides a parameter-independent figure of merit for the structural fluctuation gain.

B. Main combinatorial theorem

Theorem 6 (Exact non-trivial loop multiplicities for the three cubic Bravais lattices). *For the three cubic Bravais lattices at a single momentum scale k_0 , using the standard primary reciprocal stars (FCC: $\{111\}$ family, 8 vectors, $N_S = 4$ pairs; BCC: $\{110\}$ family, 12 vectors, $N_S = 6$ pairs; SC: $\{100\}$ family, 6 vectors, $N_S = 3$ pairs), the non-trivial 4-wave resonance-loop counts and the dimensionless geometric invariants are*

$$N_{\text{loop}}^{\text{SC}} = 0, \quad N_S^{\text{SC}} = 3, \quad \mathcal{C}_{\text{SC}} = 0, \quad (6)$$

$$N_{\text{loop}}^{\text{FCC}} = 2, \quad N_S^{\text{FCC}} = 4, \quad \mathcal{C}_{\text{FCC}} = \frac{1}{8} \approx 0.125, \quad (7)$$

$$N_{\text{loop}}^{\text{BCC}} = 6, \quad N_S^{\text{BCC}} = 6, \quad \mathcal{C}_{\text{BCC}} = \frac{1}{6} \approx 0.167. \quad (8)$$

The strict hierarchy $\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}$ is a rational topological invariant: it depends only on the combinatorial graph structure of the reciprocal star and is independent of r , κ , u , k_0 , or any physical parameter. The ratios $\mathcal{C}_{\text{BCC}}/\mathcal{C}_{\text{FCC}} = 4/3$ and the absolute loop-count ratio $N_{\text{loop}}^{\text{BCC}}/N_{\text{loop}}^{\text{FCC}} = 3$ quantify the BCC preference at the level of the combinatorial invariant.

Proof outline. Two independent methods give the same counts: direct enumeration under O_h orbit decomposition, and Burnside's lemma applied to the action of O_h on 4-element subsets of the half-star.

SC ($N_S = 3$). The half-star is $\{\pm k_0 \hat{\mathbf{e}}_x, \pm k_0 \hat{\mathbf{e}}_y, \pm k_0 \hat{\mathbf{e}}_z\}$. The orthogonality of the Cartesian basis forbids any $\sum n_j \hat{\mathbf{e}}_j = 0$ with $n_j \in \{-1, 0, +1\}$ and four non-zero entries, since this would require a repeated entry constituting an antipodal pair, violating condition (ii). Hence $N_{\text{loop}}^{\text{SC}} = 0$ exactly.

FCC ($N_S = 4$, $\{111\}$ family). Direct enumeration over the $\binom{8}{4} = 70$ four-subsets yields exactly two non-trivial unordered subsets, related by the inversion $-I \in O_h$: $\mathcal{L}_1 = \{(1, 1, 1), (1, -1, -1), (-1, 1, -1), (-1, -1, 1)\}$ and $\mathcal{L}_2 = -\mathcal{L}_1$ (in units of $k_0/\sqrt{3}$).

BCC ($N_S = 6$, $\{110\}$ family). Burnside's lemma on the O_h action over momentum-conserving 4-subsets of $\{\pm \mathbf{e}_i\}_{i=1}^6$ gives $N_{\text{orbits}} = (1/|O_h|) \sum_{g \in O_h} |\mathcal{F}(g)| = 6$. The six distinct loops are exhaustively enumerated by solving the system $s_a \mathbf{e}_{n_a} + s_b \mathbf{e}_{n_b} + s_c \mathbf{e}_{n_c} + s_d \mathbf{e}_{n_d} = 0$ with $s_i \in \{\pm 1\}$ component by component; each loop has no antipodal pair.

Computing the invariants gives $\mathcal{C}_{\text{SC}} = 0/9 = 0$, $\mathcal{C}_{\text{FCC}} = 2/16 = 1/8$, $\mathcal{C}_{\text{BCC}} = 6/36 = 1/6$. The hierarchy is an exact rational relation. $\square$

Remark 7 (Exclusion of non-cubic and quasicrystalline structures). Non-cubic Bravais structures (orthorhombic, monoclinic, triclinic) split the active modes into at least two distinct momentum magnitudes $k_0 \pm \delta q$ with $\delta q > 0$, incurring a kinetic gradient penalty $\kappa(\delta q)^2 k_0^2 \cdot V_{\text{system}}$ that is extensive in the system volume. The icosahedral quasicrystal ($N = 15$ half-vectors with Y_h symmetry) requires irrational combinations of golden-ratio integers $\{1, \tau\}$ in the dot products between half-star vectors and any closed loop; the resulting incommensurability produces a topological phase mismatch $\delta q_{\text{QC}} = k_0(\tau - \lfloor \tau \rfloor) \approx 0.618 k_0$, which translates to an extensive penalty $\Delta E_{\text{QC}} \geq \kappa(0.618 k_0)^2 k_0^2 V$. HCP is non-Bravais (two-point basis) and cannot satisfy the sub-lattice phase-matching condition without breaking the equal-amplitude O_h ansatz at $\mathcal{O}(A^2)$, introducing an inter-sublattice phase-frustration penalty.

VI. GLOBAL 12-STAR OPTIMALITY THEOREM

The combinatorial Theorem 6 compares BCC against the discrete set of cubic Bravais lattices. The following theorem strengthens the BCC selection in a qualitatively different sense: it proves that the BCC $\{110\}$ cuboctahedral star uniquely maximises a quadratic-form objective among all antipodal 12-vector subsets of the unit sphere, including continuous deformations.

Definition 8 (Pair-sum multiplicity and second moment). Let $S \subset \mathbb{S}^2$ be an antipodal set of N unit vectors. For each $v \in \mathbb{R}^3$, let $r(v)$ denote the number of ordered pairs $(\mathbf{k}_i, \mathbf{k}_j)$ with $\mathbf{k}_i, \mathbf{k}_j \in S$ and $\mathbf{k}_i + \mathbf{k}_j = v$. The *second-moment functional* is

$$\Lambda(S) \equiv \sum_{v \in \mathbb{R}^3} r(v)^2. \quad (9)$$

This functional encodes the resonance-saturation density of the configuration S .

Theorem 9 (Global 12-star optimality). *Let $S \subset \mathbb{S}^2$ be any antipodal set of $N = 12$ unit vectors. Then*

$$\Lambda(S) \leq 540, \quad (10)$$

with equality if and only if S is congruent to the BCC $\{110\}$ cuboctahedral star:

$$\Lambda(S) = 540 \iff S = O \cdot S_{\text{BCC}} \text{ for some } O \in O(3). \quad (11)$$

Proof outline. The proof proceeds in three steps: (i) decompose Λ into a diagonal contribution $r(\mathbf{0})^2 = N^2 = 144$ plus an off-diagonal sum $\sum_{v \neq 0} r(v)^2$ subject to the sum constraint $\sum_{v \neq 0} r(v) = N(N-1) = 132$; (ii) establish three sharp constraints on $r(v)$ for $v \neq 0$ — the fiber-circle bound $r(v) \leq |S \cap \mathcal{F}(v)|$, the parity constraint that $r(v)$ is even (from antipodal pairing), and the maximum-occupancy bound $r_{\max} = 4$ (from the cuboctahedral rigidity argument); (iii) solve the resulting integer optimisation $4n_4 + 2n_2 = 132$ subject to the geometric constraint $n_4 \leq 18$, giving the maximum $\sum_{v \neq 0} r(v)^2 \leq 396$ and $\Lambda(S) \leq 144 + 396 = 540$. The BCC $\{110\}$ star saturates all three constraints simultaneously: $r_{\max} = 4$, $n_4 = 18$, $n_2 = 24$, $n_1 = 12$ (explicit enumeration). Any configuration achieving $\Lambda = 540$ must place its 18 maximally resonant v -vectors on the 18 edge-midpoint directions of a cuboctahedron, uniquely determining $S = O \cdot S_{\text{BCC}}$ up to rotation by Gram-matrix uniqueness. $\square$

Remark 10 (Quantitative comparison with the icosahedral 12-star). The icosahedral 12-star (Y_h symmetry, $N = 12$) achieves $\Lambda(I_h) = 132$, less than a quarter of the BCC value: $\Lambda_{\text{BCC}}/\Lambda_{I_h} =$

$540/132 \approx 4.1$. This provides an independent and quantitative proof that quasicrystalline icosahedral order cannot compete with BCC in the single-shell $\mathbb{Z}_2$ -symmetric Brazovskii framework, complementing the topological-incommensurability argument of Remark 7.

Remark 11 (Strengthening relative to Theorem 6). Theorem 6 compares BCC against the discrete set of cubic Bravais lattices. Theorem 9 proves BCC uniqueness within the much larger continuous space of all antipodal 12-vector configurations. The result therefore upgrades the cubic Bravais ranking to a global statement about 12-star optimality on the sphere.

VII. MULTI-SHELL BCC PERSISTENCE THEOREM

The single-shell theorems above establish BCC selection at the combinatorial and global-12-star levels within one momentum scale. The following theorem extends this to the two-shell setting under explicit dominance hypotheses.

Theorem 12 (Multi-shell BCC persistence). *Consider the two-shell $\mathbb{Z}_2$ -symmetric Brazovskii free energy*

$$\mathcal{F}_{2\text{sh}} = r_0 A^2 + r_1 B^2 + \frac{u}{2} [\beta_S A^4 + \beta_V B^4 + 2\beta_{SV}(\rho) A^2 B^2], \quad (12)$$

with $r_0 < 0 < r_1$ (primary shell soft, secondary shell massive), amplitudes A and B on the primary and secondary shells, and β_S , β_V , $\beta_{SV}(\rho)$ the intra-shell and cross-shell coupling coefficients depending on the shell-radius ratio $\rho \equiv k_1/k_0$. Define the dominance parameter $\kappa \equiv |r_1|/|r_0|$.

(i) Strong primary-shell dominance, $\kappa \gg 1$. The secondary-shell amplitude obeys $B^* \rightarrow 0$ at the saddle, the free energy reduces to the single-shell form $\mathcal{F} \approx r_0 A^2 + (u/2)\beta_S A^4$, and the BCC $> \text{FCC} > \text{SC}$ hierarchy of Theorem 6 is preserved exactly, with corrections to the selection threshold suppressed by $\mathcal{O}(\kappa^{-1})$.

(ii) Competitive two-shell regime, $\kappa = \mathcal{O}(1)$. BCC remains the selected primary-shell structure as long as the cross-shell coupling satisfies the perturbative-dominance condition

$$\beta_{SV}(\rho) < \sqrt{\beta_S \cdot \beta_V}. \quad (13)$$

(iii) Secondary-shell dominance, $\kappa \ll 1$. The single-shell BCC analysis applies to shell 2, with BCC selected among secondary-shell candidates by the same combinatorial mechanism.

Proof of (i). In the limit $\kappa \gg 1$, expanding the saddle-point equations of Eq. (12) for $|r_1| \gg |r_0|$ gives $B^{*2} \approx r_0 \beta_{SV} / (-r_1 \beta_V) \rightarrow 0$. The free energy reduces to $\mathcal{F} \approx r_0 A^2 + (u/2)\beta_S A^4$, which is

precisely the single-shell problem in Eq. (1) with the BCC ranking established by Theorem 1 and Theorem 6. Corrections to the selection threshold are $\mathcal{O}(\kappa^{-1})$. $\square$

Corollary 13 (BCC selection at the TECT operating point under the strong-dominance hypothesis). *At the TECT operating point, the secondary-shell mass is parametrically larger than the primary-shell mass-squared deviation from the Brazovskii instability, so $\kappa \gg 1$ holds. By Theorem 12 (i), the single-shell BCC ranking of Theorem 1 and Theorem 6 is preserved exactly with $\mathcal{O}(\kappa^{-1})$ corrections at the TECT operating point.*

VIII. DISCUSSION

The present results show that, within the restricted single-shell SMA model at the TECT operating point, the BCC ansatz is the lowest-ranked member of the tested competitor set, with a finite ranking gap $\Delta F \sim 0.30$ (TECT code units) to the nearest competitor (gyroid). This is a non-marginal preference within the present truncation. The exact combinatorial theorem (Theorem 6) elevates the cubic Bravais ranking from a numerical to a rational topological invariant of the reciprocal star, with $\mathcal{C}_{\text{BCC}}/\mathcal{C}_{\text{FCC}} = 4/3$ (absolute loop-count ratio 3 : 1). The multi-mode robustness proposition (Proposition 3) shows that the single-shell ranking is the dominant contribution in the small-detuning regime $\delta k \ll k_0$. The global 12-star optimality theorem (Theorem 9) strengthens the result by proving that the BCC {110} star uniquely maximises the second-moment functional Λ among *all* antipodal 12-vector configurations on the unit sphere, not just the cubic Bravais subset. The multi-shell persistence theorem (Theorem 12) preserves the hierarchy in the strong primary-shell-dominance regime, which obtains at the TECT operating point per Corollary 13.

a. Honest scope. The four results above do not constitute a closed global-minimum theorem on the unrestricted configuration space. Such a closure would additionally require: (a) beyond-single-scale control extending the multi-shell analysis from two shells to general shell hierarchies; (b) beyond-mean-field control accounting for non-Gaussian fluctuation corrections to the effective coupling; (c) extension of the global-12-star result to general N -vector antipodal sets with N not equal to 12. Each of these is an open programme. Within the layered framework above, the BCC selection is rigorous at the level of the combinatorial theorem, robust against small-detuning multi-mode corrections, globally optimal on the 12-vector antipodal-set space, and persistent in the strong-shell-dominance regime of the two-shell extension; the canonical TECT bookkeeping records the result as a *conditional single-shell ranking with multi-shell persistence and global-12-star optimality*, at the canonical Pillar tier recorded in `Docs/status/TOE-FACT-SHEET.md`.

b. Implications for TECT. At the canonical Pillar tier, the present results upgrade the “why BCC?” programme from a postulate to a structural-selection candidate certified at four independent levels (combinatorial ranking, multi-mode robustness, global 12-star optimality, multi-shell persistence). The remaining beyond-mean-field and beyond-12-vector items are open programme statements rather than unverified assumptions.

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